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The Effect of Rotatory Inertia and of Shear Deformation on the Frequency and Normal Mode Equations of Uniform Beams With Simple End Conditions

New frequency and normal mode equations for flexural vibrations of six common types of simple, finite beams are presented. The derivation includes the effect of rotatory inertia and transverse-shear deformation. A specific example is given.

THE classical one-dimensional Bernoulli-Euler theory of flexural motions of elastic beams has been known to be inadequate for the vibration of higher modes. It is also inadequate for those beams when the effect of the cross-sectional dimensions on frequencies cannot be neglected. Rayleigh [1]¹ introduced the effect of rotatory inertia and Timoshenko [2, 3] extended it to include the effect of transverse-shear deformation. On the other hand, the exact equations, due to Pochhammer [4] and to Chree [5], have been derived from the general equations of the theory of elasticity, and the resulting frequency equation for flexural vibrations is discussed by Bancroft [6] and the necessary computations are carried out by Hudson [7]. Davis [8] shows that the results from Timoshenko's equation are in remarkably good agreement with those obtained by Hudson from the exact elasticity equations.

Since then there has been considerable research interest in applying the Timoshenko theory to the transient responses of beams as well as the free and forced vibrations. The present paper deals with the frequency equations and normal modes of free flexural vibrations of finite beams including the effect of shear and rotatory inertia for various cases of simple beams. Anderson [9] and Dolph [10], in dealing with this problem, give general solutions and complete analysis of uniform hinged-hinged beam. Using methods of Ritz and Galerkin, Huang [11] also presents the results for a hinged-hinged beam. Dolph, in addition, includes the analog-computer solution for the free-free beam due to Howe, et al. [12]. Earlier, Kruszewski [13] obtained frequency equations for cantilever and free-free beams by solving a complete differential equation in deflection with prescribed nonhomogeneous boundary conditions. Some of these conditions are lengthy and complicated and make the problem not easy to deal with. It is these nonhomogeneous boundary conditions which limit his solutions only to the two types of beams mentioned. In the present paper a somewhat different approach is used. The novel features are (a) the solutions are obtained for two complete differential equations in total deflection and bending

slope, respectively, (b) the constants in these solutions are related by any one of the two original coupled equations from which the foregoing two complete differential equations are derived, and (c) the boundary conditions prescribed are homogeneous. The frequency and normal mode equations for all six common types of simple, finite beams are obtained. A numerical example is given.

Differential Equations and Boundary Conditions

The coupled equations for the total deflection y and the bending slope ψ are given by Timoshenko [14] as

$$EI \frac{\partial^2 \psi}{\partial x^2} + k \left(\frac{\partial y}{\partial x} - \psi \right) AG - \frac{I\gamma}{g} \frac{\partial^2 \psi}{\partial t^2} = 0 \quad (1)$$

$$\frac{\gamma A}{g} \frac{\partial^2 y}{\partial t^2} - k \left(\frac{\partial^2 y}{\partial x^2} - \frac{\partial \psi}{\partial x} \right) AG = 0 \quad (2)$$

in which

- E = modulus of elasticity
- G = modulus of rigidity
- I = area moment of inertia of cross section
- A = cross-sectional area
- γ = weight per unit volume
- k = numerical shape factor for cross section

Eliminating ψ or y from equations (1) and (2), the following two complete differential equations in y and ψ are obtained:

$$EI \frac{\partial^4 y}{\partial x^4} + \frac{\gamma A}{g} \frac{\partial^2 y}{\partial t^2} - \left(\frac{\gamma I}{g} + \frac{EI}{gk} \frac{\gamma}{G} \right) \frac{\partial^4 y}{\partial x^2 \partial t^2} + \frac{\gamma I}{g} \frac{\gamma}{gkG} \frac{\partial^4 y}{\partial t^4} = 0 \quad (3)$$

$$EI \frac{\partial^4 \psi}{\partial x^4} + \frac{\gamma A}{g} \frac{\partial^2 \psi}{\partial t^2} - \left(\frac{\gamma I}{g} + \frac{EI}{gk} \frac{\gamma}{G} \right) \frac{\partial^4 \psi}{\partial x^2 \partial t^2} + \frac{\gamma I}{g} \frac{\gamma}{gkG} \frac{\partial^4 \psi}{\partial t^4} = 0 \quad (4)$$

Let

$$y = Y e^{i p t} \quad (5)$$

$$\psi = \Psi e^{i p t} \quad (6)$$

$$\xi = x/L \quad (7)$$

¹ Numbers in brackets designate References at end of paper.

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where

Y = normal function of y
 Ψ = normal function of ψ
 ξ = nondimensional length of beam
 $j = \sqrt{-1}$
 p = angular frequency
 L = length of beam

Omitting the factor e^{jpt} , equations (1) to (4) are reduced to

$$s^2 \Psi'' - (1 - b^2 r^2 s^2) \Psi + Y'/L = 0 \quad (8)$$

$$Y'' + b^2 s^2 Y - L \Psi' = 0 \quad (9)$$

$$Y^{iv} + b^2(r^2 + s^2) Y'' - b^2(1 - b^2 r^2 s^2) Y = 0 \quad (10)$$

$$\Psi^{iv} + b^2(r^2 + s^2) \Psi'' - b^2(1 - b^2 r^2 s^2) \Psi = 0 \quad (11)$$

where

$$b^2 = \frac{1}{EI} \frac{\gamma A}{g} L^4 p^2 \quad (12)$$

$$r^2 = \frac{I}{AL^2} \quad (13)$$

$$s^2 = \frac{EI}{kAGL^2} \quad (14)$$

and the primes for Y and Ψ represent differentiation with respect to ξ .

The necessary and sufficient boundary conditions for the beams are found as follows:

Hinged end

$$Y = 0 \quad (15)$$

$$\Psi' = 0 \quad (16)$$

Clamped end

$$Y = 0 \quad (17)$$

$$\Psi = 0 \quad (18)$$

Free end

$$\Psi' = 0 \quad (19)$$

$$\frac{1}{L} Y' - \Psi = 0 \quad (20)$$

Solutions

The solutions of (10) and (11) can be found as

$$Y = C_1 \cosh b\alpha\xi + C_2 \sinh b\alpha\xi + C_3 \cos b\beta\xi + C_4 \sin b\beta\xi \quad (21)$$

$$\Psi = C_1' \sinh b\alpha\xi + C_2' \cosh b\alpha\xi + C_3' \sin b\beta\xi + C_4' \cos b\beta\xi \quad (22)$$

where

$$\alpha = \frac{1}{\sqrt{2}} \{ \mp(r^2 + s^2) + [(r^2 - s^2)^2 + 4/b^2]^{1/2} \}^{1/2} \quad (23)$$

and

$$[(r^2 - s^2)^2 + 4/b^2]^{1/2} > (r^2 + s^2)$$

is assumed.

In case

$$[(r^2 - s^2)^2 + 4/b^2]^{1/2} < (r^2 + s^2),$$

we write

$$\alpha = \frac{1}{\sqrt{2}} \{ (r^2 + s^2) - [(r^2 - s^2)^2 + 4/b^2]^{1/2} \}^{1/2} j = j\alpha' \quad (24)$$

then (21) and (22) are replaced by

$$Y = C_1 \cos b\alpha'\xi + jC_2 \sin b\alpha'\xi + C_3 \cos b\beta\xi + C_4 \sin b\beta\xi \quad (25)$$

$$\Psi = jC_1' \sin b\alpha'\xi + C_2' \cos b\alpha'\xi + C_3' \sin b\beta\xi + C_4' \cos b\beta\xi \quad (26)$$

Solutions of (21-22), or (25-26) are naturally the solutions of the original coupled equations (8) and (9).

Only one half of the constants in equations (21) and (22) are independent. They are related by the equations (8) or (9) as follows:

$$C_1 = \frac{L}{b\alpha} [1 - b^2 s^2 (\alpha^2 + r^2)] C_1' \quad (27)$$

$$C_2 = \frac{L}{b\alpha} [1 - b^2 s^2 (\alpha^2 + r^2)] C_2' \quad (28)$$

$$C_3 = -\frac{L}{b\beta} [1 + b^2 s^2 (\beta^2 - r^2)] C_3' \quad (29)$$

$$C_4 = \frac{L}{b\beta} [1 + b^2 s^2 (\beta^2 - r^2)] C_4' \quad (30)$$

or

$$C_1' = \frac{b}{L} \frac{\alpha^2 + s^2}{\alpha} C_1 \quad (31)$$

$$C_2' = \frac{b}{L} \frac{\alpha^2 + s^2}{\alpha} C_2 \quad (32)$$

$$C_3' = -\frac{b}{L} \frac{\beta^2 - s^2}{\beta} C_3 \quad (33)$$

$$C_4' = \frac{b}{L} \frac{\beta^2 - s^2}{\beta} C_4 \quad (34)$$

Frequency Equations

The application of appropriate boundary conditions (15-20) and relations of integration constants (27-34) to equations (21) and (22) yields for each type of beam a set of four homogeneous linear algebraic equations in four constants C_1 to C_4 with or without primes. In order that the solutions other than zero may exist the determinant of the coefficients of C_i must be equal to zero. This leads to the frequency equation in each case from which the natural frequencies can be determined.

In the following, six common types of beams will be identified by a compound adjective which describes the end conditions at $\xi = 0$ and $\xi = 1$ or $x = 0$ and $x = L$. They are (a) supported-supported or hinged-hinged, (b) free-free, (c) clamped-clamped, (d) clamped-free, (e) clamped-supported, and (f) supported-free.

The frequency equations thus obtained are as follows:

(a) Supported-supported beam

$$\sin b\beta = 0 \quad (35)$$

(b) Free-free beam

$$2 - 2 \cosh b\alpha \cos b\beta + \frac{b}{(1 - b^2 r^2 s^2)^{1/2}} [b^2 s^2 (r^2 - s^2)^2 + (3r^2 - s^2)] \sinh b\alpha \sin b\beta = 0 \quad (36)$$

(c) Clamped-clamped beam

$$2 - 2 \cosh b\alpha \cos b\beta + \frac{b}{(1 - b^2 r^2 s^2)^{1/2}} [b^2 s^2 (r^2 - s^2)^2 + (3s^2 - r^2)] \sinh b\alpha \sin b\beta = 0 \quad (37)$$

(d) Clamped-free beam

$$2 + [b^2 (r^2 - s^2)^2 + 2] \cosh b\alpha \cos b\beta - \frac{b(r^2 + s^2)}{(1 - b^2 r^2 s^2)^{1/2}} \sinh b\alpha \sin b\beta = 0 \quad (38)$$

(e) Clamped-supported beam

$$\lambda \zeta \tanh b\alpha - \tan b\beta = 0 \quad (39)$$

(f) Supported-free beam

$$\lambda \tanh b\alpha - \zeta \tan b\beta = 0 \quad (40)$$

in which

$$\lambda = \alpha/\beta \quad (41)$$

$$\begin{aligned} \zeta &= (\alpha^2 + r^2)/(\alpha^2 + s^2) = (\beta^2 - s^2)/(\beta^2 - r^2) \\ &= (\alpha^2 + r^2)/(\beta^2 - r^2) = (\beta^2 - s^2)/(\alpha^2 + s^2) \end{aligned} \quad (42)$$

When

$$[(r^2 - s^2)^2 + 4/b^2]^{1/2} < (r^2 + s^2), \quad b^2 r^2 s^2 > 1$$

correspondingly. It is convenient to use $\alpha = j\alpha'$ and

$$(1 - b^2 r^2 s^2)^{1/2} = j(b^2 r^2 s^2 - 1)^{1/2} \quad (43)$$

$$\lambda = j\lambda' \quad (44)$$

where $\lambda' = \alpha'/\beta$ and transform the frequency equations (31-36) for the case

$$[(r^2 - s^2)^2 + 4/b^2]^{1/2} > (r^2 + s^2)$$

to those for the case

$$[(r^2 - s^2)^2 + 4/b^2]^{1/2} < (r^2 + s^2)$$

Thus we obtain the frequency equations as follows:

(a) Supported-supported beam

$$\sin b\beta = 0 \quad (45)$$

(b) Free-free beam

$$2 - 2 \cos b\alpha' \cos b\beta + \frac{b}{(b^2 r^2 s^2 - 1)^{1/2}} [b^2 r^2 (r^2 - s^2)^2 + (3r^2 - s^2)] \sin b\alpha' \sin b\beta = 0 \quad (46)$$

(c) Clamped-clamped beam

$$2 - 2 \cos b\alpha' \cos b\beta + \frac{b}{(b^2 r^2 s^2 - 1)^{1/2}} [b^2 s^2 (r^2 - s^2)^2 + (3s^2 - r^2)] \sin b\alpha' \sin b\beta = 0 \quad (47)$$

(d) Clamped-free beam

$$2 + [b^2 (r^2 - s^2)^2 + 2] \cos b\alpha' \cos b\beta - \frac{b(r^2 + s^2)}{(b^2 r^2 s^2 - 1)^{1/2}} \sin b\alpha' \sin b\beta = 0 \quad (48)$$

(e) Clamped-supported beam

$$\lambda' \zeta \tan b\alpha' + \tan b\beta = 0 \quad (49)$$

(f) Supported-free beam

$$\lambda' \tan b\alpha' + \zeta \tan b\beta = 0 \quad (50)$$

Normal Modes

For each type of beam the roots of the frequency equations, b_i , $i = 1, 2, 3, \dots$, give the eigenvalues of the problem. The corresponding eigenfunctions, normal modes Y_i and Ψ_i , can be obtained accordingly. In the following list of normal modes the subscript i is omitted for Y , Ψ , b , α , β , and the constants D and H . Since the coefficients in Y and Ψ are related, the constants D and H are connected through any one of the equations of (27-30) or (31-34).

The normal modes Y and Ψ for various simple beams are as follows: In case there are two pairs of Y and Ψ the first pair is for the case when

$$[(r^2 - s^2)^2 + 4/b^2]^{1/2} > (r^2 + s^2)$$

and the second pair,

$$[(r^2 - s^2)^2 + 4/b^2]^{1/2} < (r^2 + s^2).$$

(a) Supported-supported beam

$$Y = D \sin b\beta \xi \quad (51)$$

$$\Psi = H \sin b\beta \xi \quad (52)$$

(b) Free-free beam

$$Y = D \left[\cosh b\alpha \xi + \lambda \delta \sinh b\alpha \xi + \frac{1}{\zeta} \cos b\beta \xi + \delta \sin b\beta \xi \right] \quad (53)$$

$$\Psi = H \left[\cosh b\alpha \xi - \frac{\delta}{\lambda} \sinh b\alpha \xi + \zeta \cos b\beta \xi + \frac{1}{\delta} \sin b\beta \xi \right] \quad (54)$$

where

$$\delta = \frac{\cosh b\alpha - \cos b\beta}{\lambda \sinh b\alpha - \zeta \sin b\beta} \quad (55)$$

and

$$Y = D \left[\cos b\alpha' \xi - \lambda' \eta \sin b\alpha' \xi + \frac{1}{\zeta} \cos b\beta \xi + \eta \sin b\beta \xi \right] \quad (56)$$

$$\Psi = H \left[\cos b\alpha' \xi - \frac{\eta}{\lambda'} \sin b\alpha' \xi + \zeta \cos b\beta \xi + \frac{1}{\eta} \sin b\beta \xi \right] \quad (57)$$

where

$$\eta = - \frac{\cos b\alpha' - \cos b\beta}{\lambda' \sin b\alpha' + \zeta \sin b\beta} \quad (58)$$

(c) Clamped-clamped beam

$$Y = D[\cosh b\alpha\xi + \lambda\zeta\delta \sinh b\alpha\xi - \cos b\beta\xi + \delta \sin b\beta\xi] \quad (59)$$

$$\Psi = H \left[\cosh b\alpha\xi + \frac{\theta}{\lambda\zeta} \sinh b\alpha\xi - \cos b\beta\xi + \theta \sin b\beta\xi \right] \quad (60)$$

where

$$\delta = \frac{-\cosh b\alpha + \cos b\beta}{\lambda\zeta \sinh b\alpha + \sin b\beta} \quad (61)$$

$$\theta = \frac{-\cosh b\alpha + \cos b\beta}{\frac{1}{\lambda\zeta} \sinh b\alpha + \sin b\beta} \quad (62)$$

and

$$Y = D[\cos b\alpha'\xi - \lambda'\zeta\eta \sin b\alpha'\xi - \cos b\beta\xi + \eta \sin b\beta\xi] \quad (63)$$

$$\Psi = H \left[\cos b\alpha'\xi + \frac{\mu}{\lambda'\zeta} \sin b\alpha'\xi - \cos b\beta\xi + \mu \sin b\beta\xi \right] \quad (64)$$

where

$$\eta = \frac{\cos b\alpha' - \cos b\beta}{\lambda'\zeta \sin b\alpha' - \sin b\beta} \quad (65)$$

$$\mu = \frac{-\cos b\alpha' + \cos b\beta}{\frac{1}{\lambda'\zeta} \sin b\alpha' + \sin b\beta} \quad (66)$$

(d) Clamped-free beam

$$Y = D[\cosh b\alpha\xi - \lambda\zeta\delta \sinh b\alpha\xi - \cos b\beta\xi + \delta \sin b\beta\xi] \quad (67)$$

$$\Psi = H \left[\cosh b\alpha\xi + \frac{\theta}{\lambda\zeta} \sinh b\alpha\xi - \cos b\beta\xi + \theta \sin b\beta\xi \right] \quad (68)$$

where

$$\delta = \frac{\frac{1}{\lambda} \sinh b\alpha - \sin b\beta}{\zeta \cosh b\alpha + \cos b\beta} \quad (69)$$

$$\theta = -\frac{\lambda \sinh b\alpha + \sin b\beta}{\frac{1}{\zeta} \cosh b\alpha + \cos b\beta} \quad (70)$$

and

$$Y = D[\cos b\alpha'\xi + \lambda'\zeta\eta \sin b\alpha'\xi - \cos b\beta\xi + \eta \sin b\beta\xi] \quad (71)$$

$$\Psi = H \left[\cos b\alpha'\xi - \frac{1}{\lambda'\zeta} \mu \sin b\alpha'\xi - \cos b\beta\xi + \mu \sin b\beta\xi \right] \quad (72)$$

where

(e) Clamped-supported beam

$$Y = D[\cosh b\alpha\xi - \coth b\alpha \sinh b\alpha\xi - \cos b\beta\xi + \cot b\beta \sin b\beta\xi] \quad (75)$$

$$\Psi = H \left[\cosh b\alpha\xi + \frac{\theta}{\lambda\zeta} \sinh b\alpha\xi - \cos b\beta\xi + \theta \sin b\beta\xi \right] \quad (76)$$

where

$$\theta = -\frac{\lambda \sinh b\alpha + \sin b\beta}{\frac{1}{\zeta} \cosh b\alpha + \cos b\beta} \quad (77)$$

and

$$Y = D[\cos b\alpha'\xi - \cot b\alpha' \sin b\alpha'\xi - \cos b\beta\xi + \cot b\beta \sin b\beta\xi] \quad (78)$$

$$\Psi = H \left[\cos b\alpha'\xi - \frac{1}{\lambda'\zeta} \mu \sin b\alpha'\xi - \cos b\beta\xi + \mu \sin b\beta\xi \right] \quad (79)$$

where

$$\mu = \frac{\lambda' \sin b\alpha' - \sin b\beta}{\frac{1}{\zeta} \cos b\alpha' + \cos b\beta} \quad (80)$$

(f) Supported-free beam

$$Y = \lambda \frac{\cos b\beta}{\cosh b\alpha} \sinh b\alpha\xi + \sin b\beta\xi \quad (81)$$

$$\Psi = \frac{1}{\lambda} \frac{\sin b\beta}{\sinh b\alpha} \cosh b\alpha\xi + \cos b\beta\xi \quad (82)$$

and

$$Y = -\lambda' \frac{\cos b\beta}{\cos b\alpha'} \sin b\alpha'\xi + \sin b\beta\xi \quad (83)$$

$$\Psi = -\frac{1}{\lambda'} \frac{\sin b\beta}{\sin b\alpha'} \cos b\alpha'\xi + \cos b\beta\xi \quad (84)$$

The condition of orthogonality of normal modes can be found [11] as

$$\int_0^1 (Y_m Y_n + r_g^2 \Psi_m \Psi_n) d\xi = 0 \quad m \neq n \quad (85)$$

for m th and n th modes, and r_g is the radius of gyration of the cross section around the principal axis normal to the plane of motion and is given as

$$r_p^2 = I/A \quad (86)$$

The normal modes would also be useful for studying the forced vibration of beams, and beam and plate vibrations by variational methods as they are used for conventional beams and plates.

Deflection, Slope, Moment, and Shear

With Y and Ψ known, the total deflection, slopes, moment, and shear can be written as follows:

Total deflection

$$y = \sum a_i Y_i e^{j p_i t + \epsilon} \quad (87)$$

Bending slope

$$\psi = \sum \bar{a}_i \Psi_i e^{j p_i t + \bar{\epsilon}} \quad (88)$$

Total slope

$$\frac{\partial y}{\partial x} = \sum a_i \frac{dY_i}{dx} e^{j p_i t + \epsilon} \quad (89)$$

Shear slope

$$\varphi = \frac{\partial y}{\partial x} - \psi \quad (90)$$

Moment

$$M = -EI \frac{\partial \psi}{\partial x} \quad (91)$$

Shear

$$Q = k \varphi A G = k \left(\frac{\partial y}{\partial x} - \psi \right) A G \quad (92)$$

The constants a_i , ϵ_i , $\bar{a}_i$, and $\bar{\epsilon}_i$ have to be evaluated from the given initial conditions; i.e., the initial values of y and ψ and their time derivatives.

Numerical Example

For a given beam with r and s known, the b_i ($i = 1, 2, 3, \dots$) can be found from the appropriate frequency equations and the corresponding p_i are then calculated by the equation (12). However, these frequency equations are highly transcendental and not to be solved simply. This difficulty is overcome by the use of frequency charts which are obtained from the solution of these transcendental frequency equations for various types of beams and various combinations of r and s .

In this paper the frequency chart is given only for a special case of clamped-free beam as an illustration. Assume $k = 2/3$, $E/G = 8/3$, then $E/kG = 4$ and $s = 2r$ for this special case. Let p_0 be the corresponding frequencies from classical theory; the graphical representation of p/p_0 versus r for the first five modes of a finite cantilever beam is presented in Fig. 1 with the range of r from 0 to 0.10. It is seen that the reduction of the ratio of natural frequencies is increased by increasing the values of r and s and for the higher modes. In the curves shown as high as 70 per cent reduction is possible.

The application of these curves of the frequency chart will now be illustrated for a steel cantilever beam 14.4 in. in length, 1 sq in. in square cross section, $E = 30 \times 10^6$ psi, and $\gamma = 0.28$ lb/in.³ Assuming $k = 2/3$, $E/G = 8/3$, we have $r = 0.02$, $s = 0.04$. Since $s = 2r$ we have from Fig. 1

$$p/p_0 = 0.985, 0.975, 0.930, 0.883, 0.835 \quad (93)$$

for the first five modes. Since

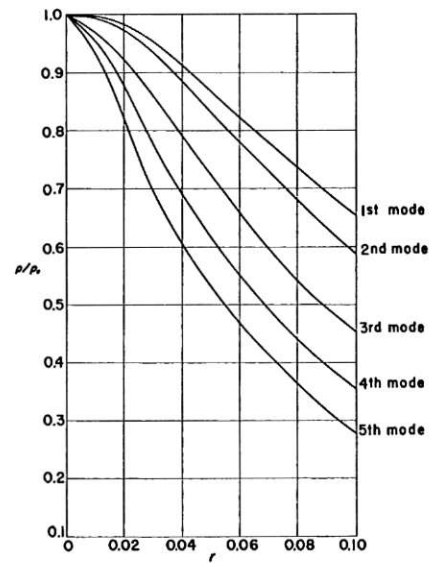


Fig. 1 Corrections in natural frequencies of a clamped-free beam owing to rotatory inertia and shear deformation

$$b/b_0 = p/p_0 \quad (94)$$

and the corresponding b_0 are

$$b_0 = 3.51, 22.0, 61.9, 121.2, 199.9 \quad (95)$$

we have

$$b = 3.46, 21.4, 57.5, 107.0, 167.0 \quad (96)$$

The p_i are then obtained from (12) as

$$p = 977, 6040, 16200, 30200, 47200 \quad (97)$$

Equations (23) give values of α_i and β_i for each b_i . The corresponding values of λ_i , ζ_i , δ_i , and θ_i are then calculated from (41-42) and (69-70.) The normal modes Y and Ψ are then given by (67-68). For the instance of foregoing example the superimposed modes of the first five normal modes of Y and Ψ are shown by the solid curves in (a) and (b) of Fig. 2, respectively. The relative magnitudes of each of these five component normal modes are so chosen that they reduce to the magnitudes according to the classical theory given by reference [15]. The corresponding superimposed modes of Y and Ψ from these reduced magnitudes are shown by the dotted curves in the same figure for comparison.

Frequency charts² for the first five modes of six common types of beams with various combinations of r and s are available.

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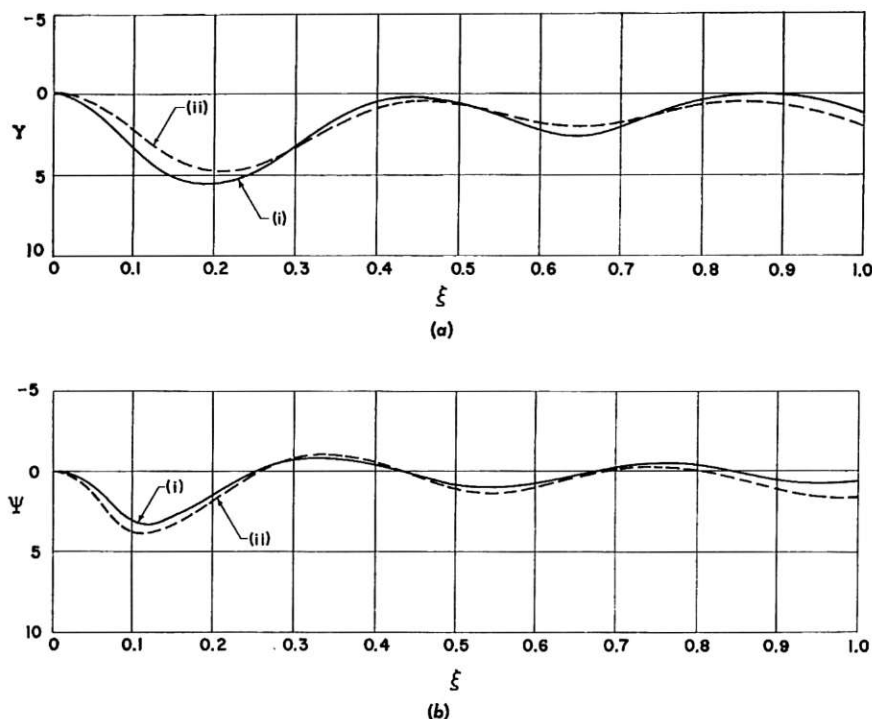


Fig. 2 Comparison of superimposed modes of five component modes each according to classical theory and according to derived relations (a) deflection, (b) bending slope (i for derived relations, ii for classical theory)

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